

Government Engineering College, Sheikhpura

B.Tech 1st Semester Mid Sem. Exam 2026

Branch: ECE

Full Marks: - 20

Paper:- Basic Electronics

Time: - 2 Hours

Instructions:

- i. The marks are indicated in the right-hand margin.
 - ii. There are 8 Questions in this Paper.
 - iii. Attempt 6 Questions in all.
 - iv. Question No. 1 is Compulsory.
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Objective:-

1 X 5 = 05

1. i. In a PNP transistor, the majority charge carriers in the emitter are:

(CO3, R)

- a) Electrons
- b) Holes
- c) Neutrons
- d) Protons

ii. What is drift and diffusion current ?

(CO1, R)

iii. Which of the following is an impurity added in a semiconductor to make it a P-Type semiconductor?

(CO1, R)

- a) Indium
- b) Phosphorus
- c) Arsenic
- d) Antimony

iv. Zener diode in reverse bias state has a controlled breakdown that allows current to keep the voltage close to which of the following?

(CO2, U)

- a) Threshold voltage
- b) Zener breakdown voltage
- c) Conduction voltage
- d) Knee voltage

v. Which of the following are known as universal gates?

(CO2, U)

- a) NAND & NOR
- b) AND & OR
- c) XOR & OR
- d) EX-NOR & XOR

Subjective Question:-

3 X5 = 15

Attempt any 5 Questions:

- Q2. Explain PN junction diode & Its Application. (CO1, U)
- Q 3. Describe Zener diode Working & VI characteristics. (CO1, U)
- Q4. a. What is the difference between basic gates and universal gates? Provide an example of how a universal gate can implement a basic gate. (CO2, Ap)
 b. Convert the Decimal number 841.354 to Binary Number. (CO2, U)
- Q5. a. Explain one's and two's complements arithmetic with example. (CO2, U)
 b. Convert the Octal number 476.62 to Decimal Number. (CO2, Ap)
 c. Convert Gray Code 10101110 to Binary Number. (CO2, Ap)
- Q6. Explain Direct and Indirect band -gap Semiconductor. (CO1,R)
- Q7. Explain Full wave rectifiers. (CO2,Ap)
- Q8. Describe the Operation of NPN transistors & Characteristics of C-B configuration. (CO3,Ap)